

OVERVIEW OF MOTILITY &

NERVOUS CONTROL &

CIRCULATION OF GIT

MOTILITY OF THE GUT

NERVES

+

**PACEMAKER
ACTIVITY**



MOTILITY

+

HORMONES

PACEMAKER OF THE GUT

It is a thin layer of interstitial cells ;
interstitial cells of Cajal, lying
between circular and longitudinal cell
layer.

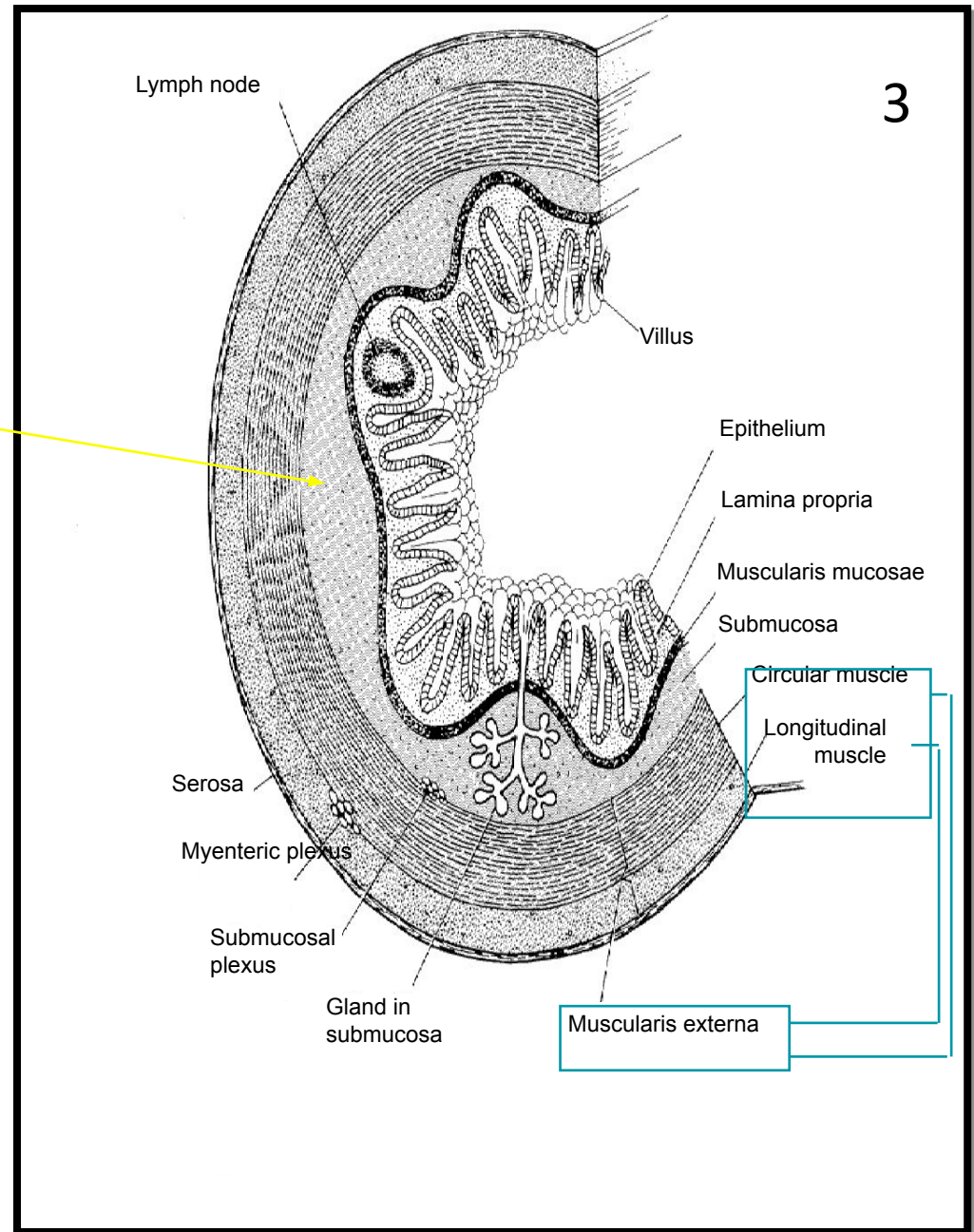
interstitial cells of Cajal

It is characterized by:

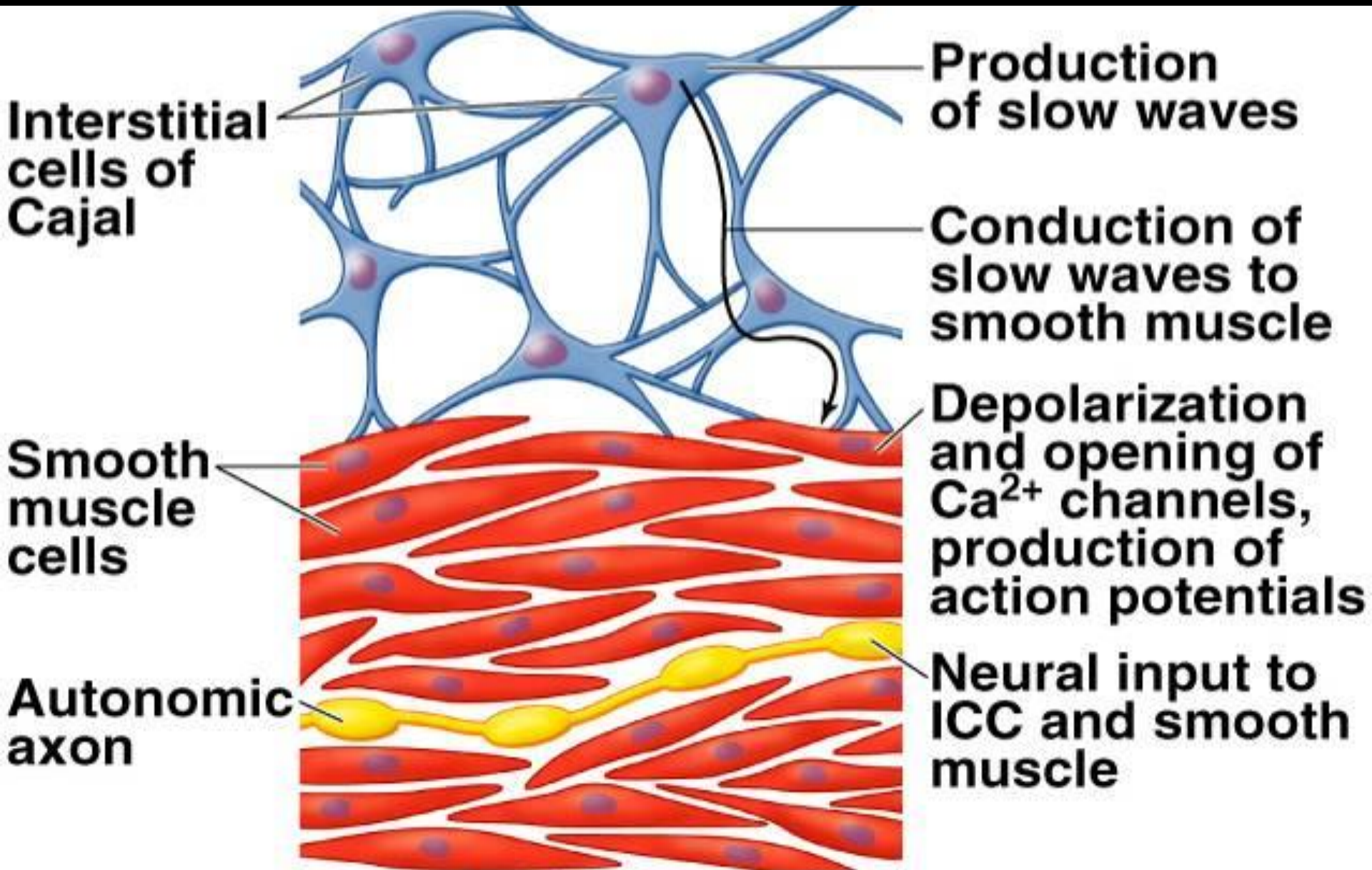
- ★ a slow wave activity
- ★ gap junctions
- ★ tight synapses with enteric nerves

Pacemaker activity:

Thin layer of
interstitial cells
(**interstitial
cells of Cajal**)
between
circular and
longitudinal cell
layer.
Conduction
through gap
junctions.



PACEMAKER OF THE GUT

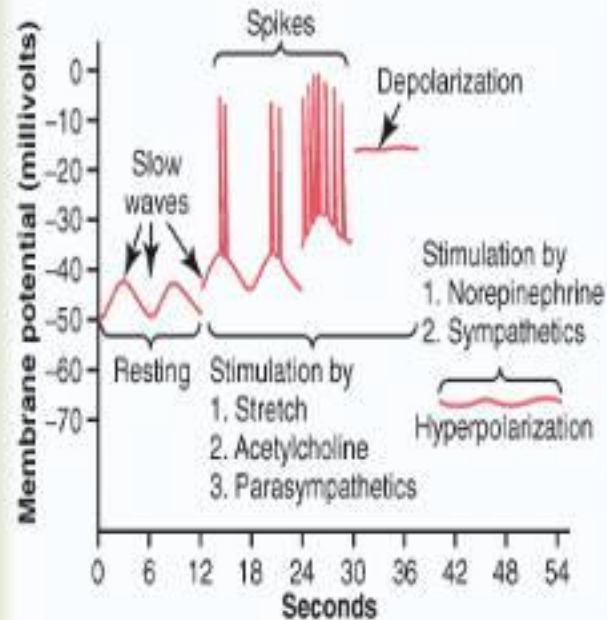


Electro-physiology of smooth muscles of the GUT

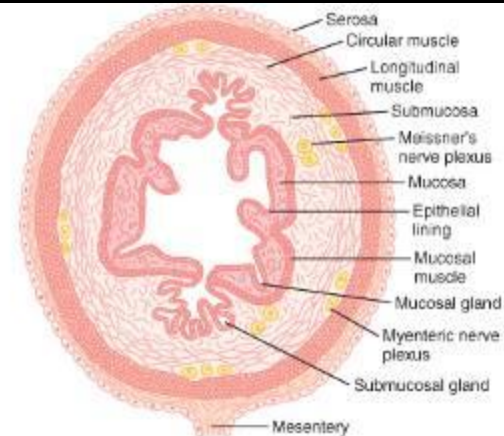
Regulation of gastric emptying

- **Gastric: promote emptying**
 - Stretch mediated in response to volume
 - Gastrin - promotes the pyloric pump
 - **Duodenal: inhibit emptying**
 - Neural
 - Intrinsic
 - Extrinsic
 - Endocrine
 - CCK
 - Secretin
 - GIP
- } Distension, irritation- acidity/osmolar change
- } Fats in chyme

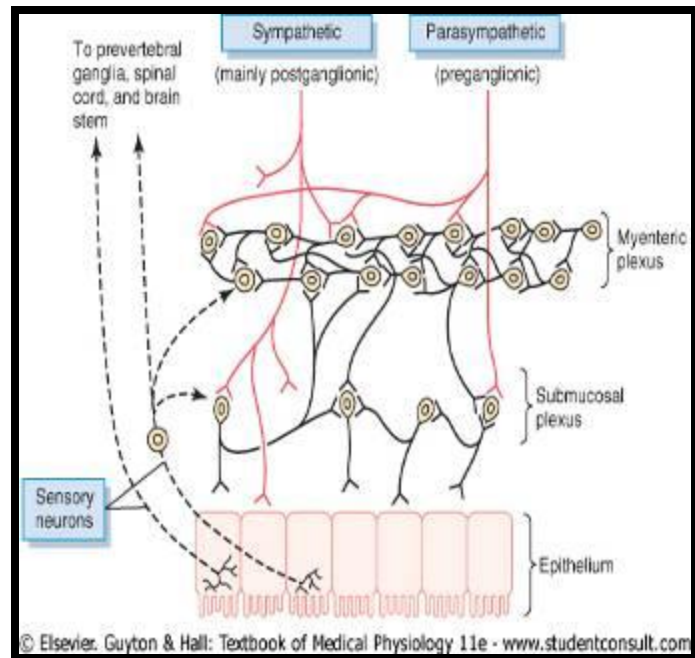
Gastrointestinal motility



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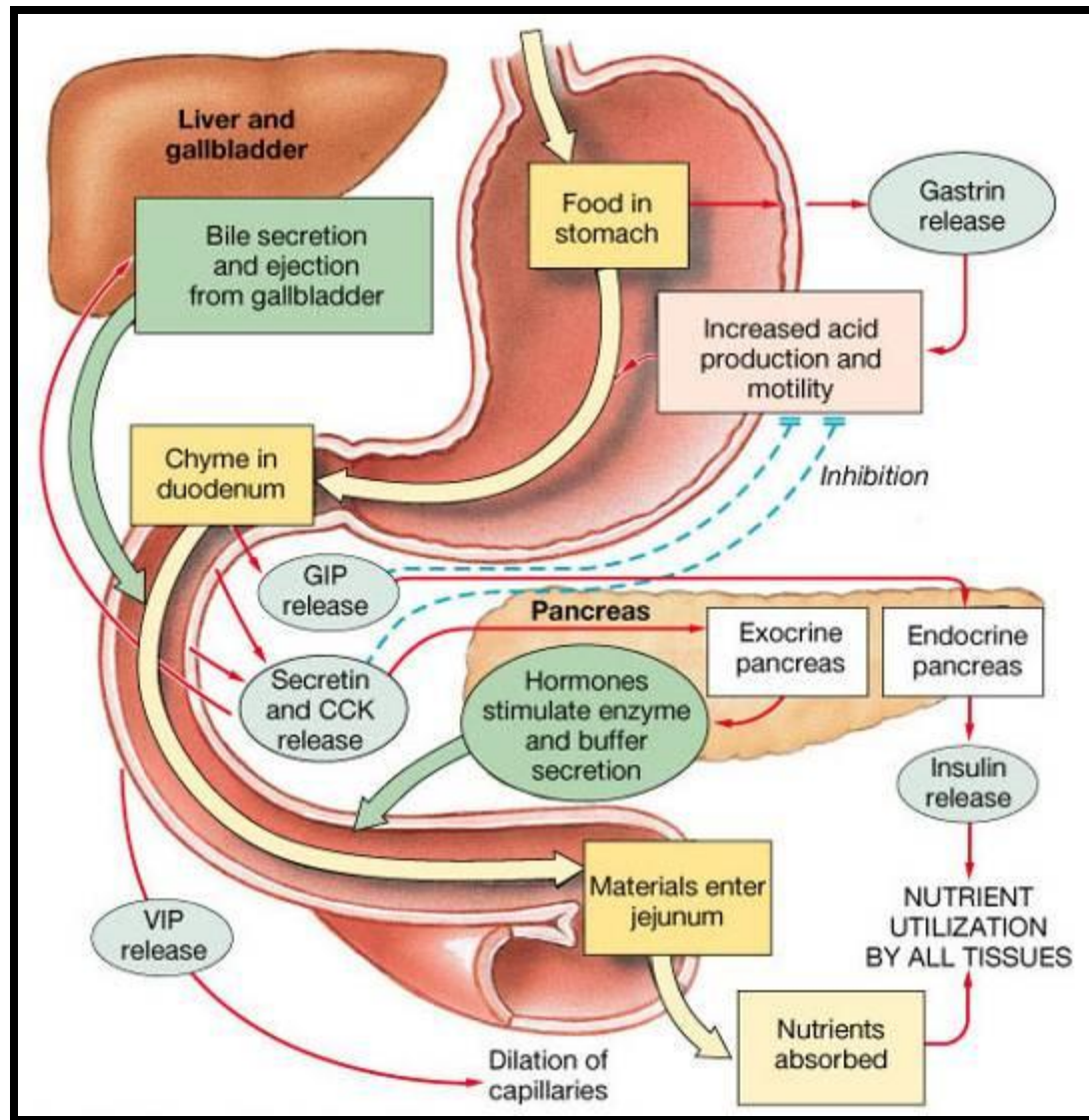


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Gastric emptying

- Intense peristaltic waves promote antral emptying
- Forces a small amount of “chyme” into the duodenum
- Pylorus acts as a pump and a sphincter
- Water and other fluids pass with ease
- Clear liquids are nearly empty within 2 hours

Overview of gastric emptying



Resting membrane potential

It is -50 to - 60 mV with potential oscillations

Slow Waves = Pacemaker Potential = Basic Electric Rhythm
(BER)

**It is spontaneous rhythmic fluctuation in
the smooth muscles membrane potential
between – 60 mv & – 40 mv.**

It is initiated by the interstitial cells of Cajal

Pacemaker potential

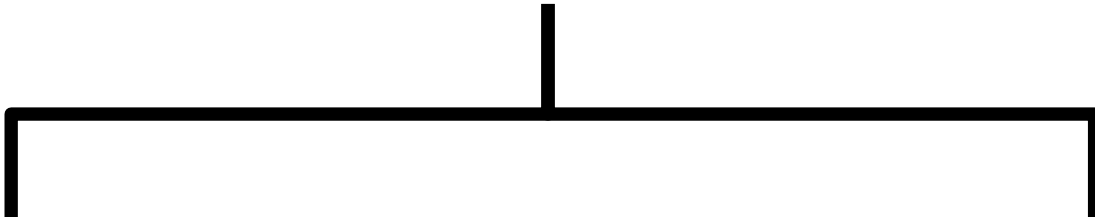
They are **not** followed by muscle contraction unless they reach a threshold for the spike potentials.

The action potentials (spike potentials)

is produced when slow-waves reach the threshold automatically due to **Opening of L-type calcium channels**

This is induced by **Stretch of smooth muscle, vagal stimulation, hormones or myenteric plexus stimulation.**

Functional types of movements

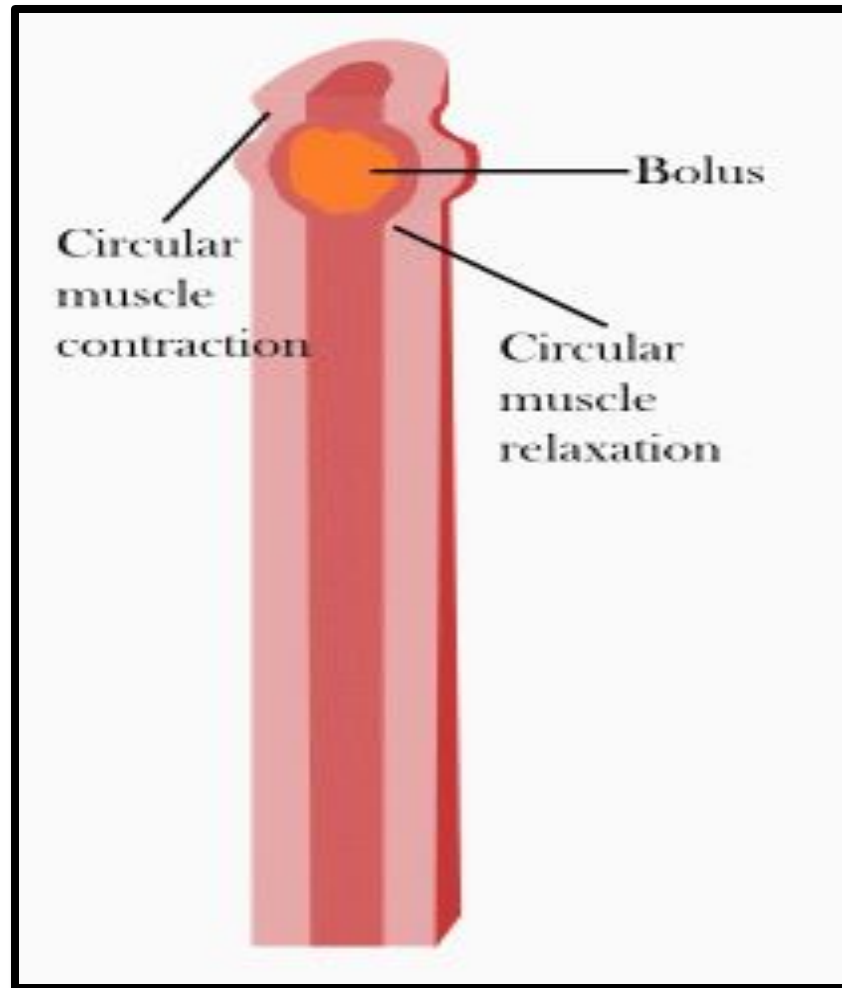


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graph TD; A[Functional types of movements] --> B[Propulsive movements (peristalsis)]; A --> C[Mixing movements];
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**Propulsive
movements
(peristalsis)**

**Mixing
movements**

Peristalsis



Peristalsis

This signal is transmitted by
interneurons

Excitatory (proximal) motor neuron

Inhibitory (distal) motor neurons,

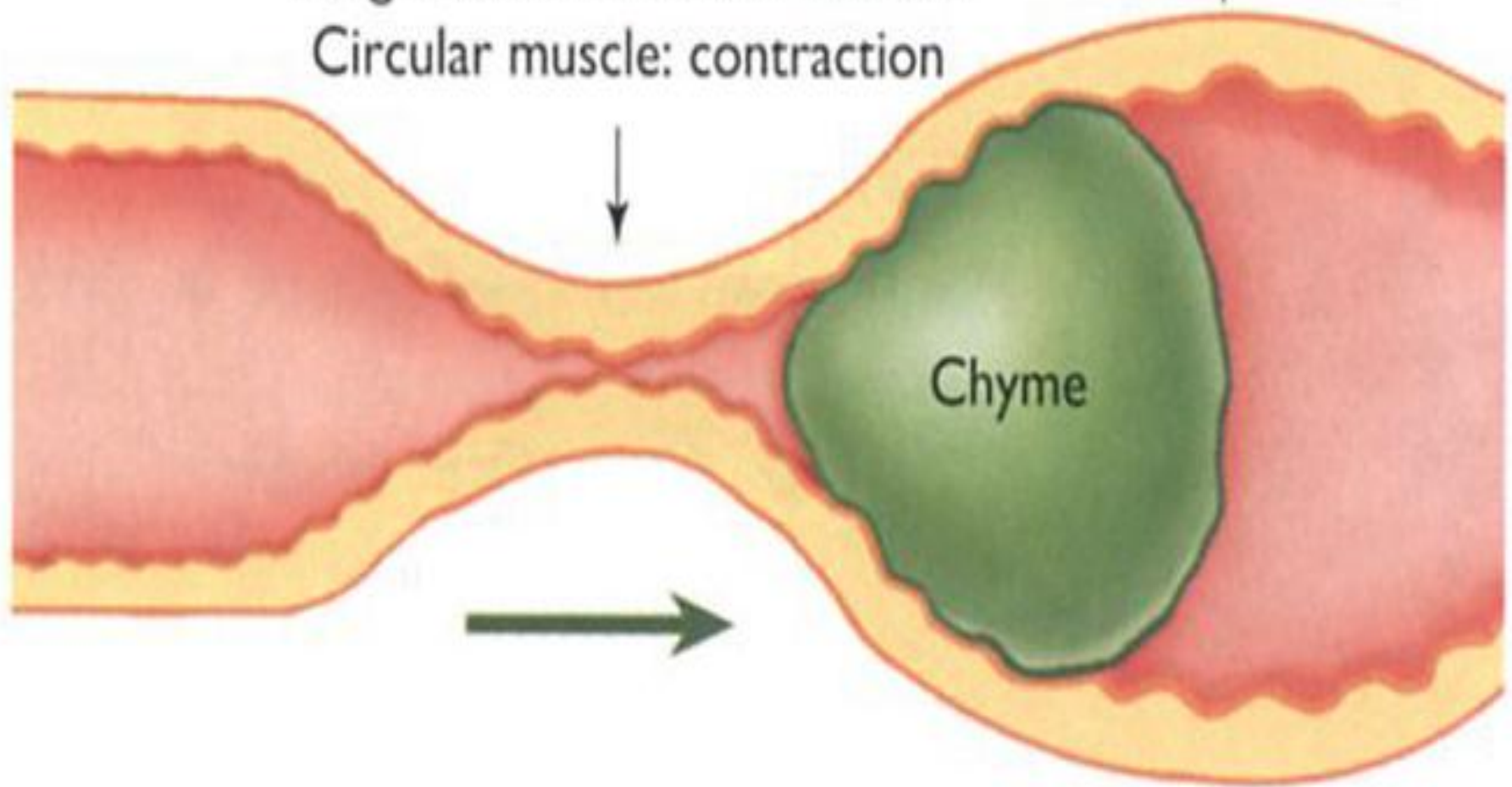
bolus is transported in oral to caudal
direction

Longitudinal muscle: contraction

Circular muscle: relaxation

Longitudinal muscle: relaxation

Circular muscle: contraction



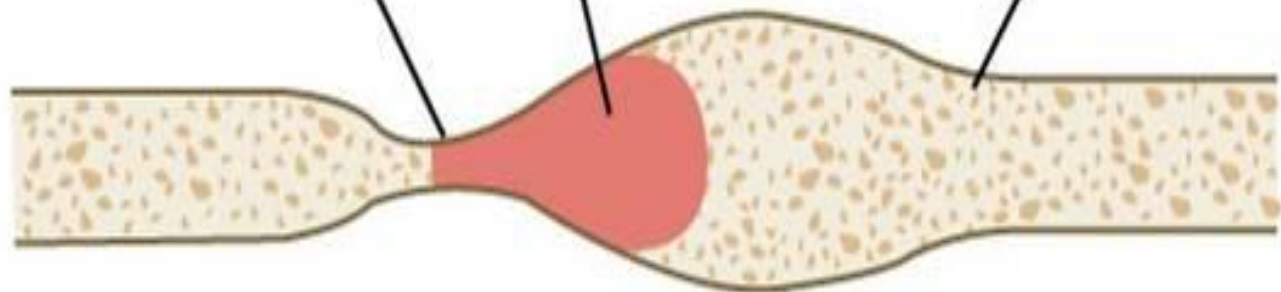
**Peristaltic contractions are responsible
for forward movement**

Contraction

Bolus

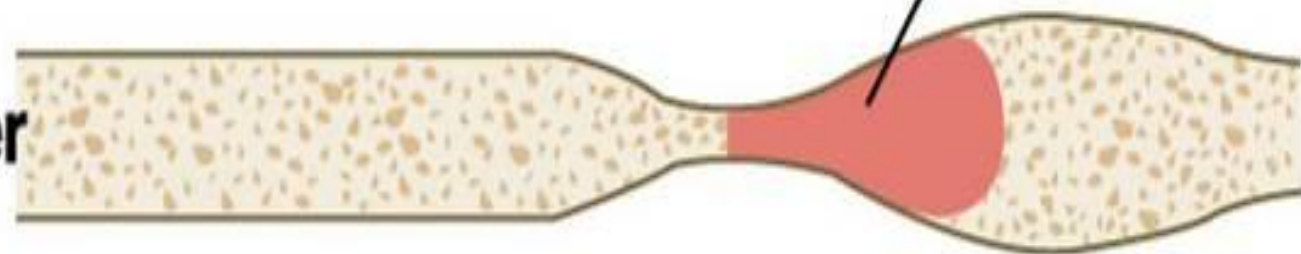
Receiving segment

Time zero



Bolus moves forward

Seconds later



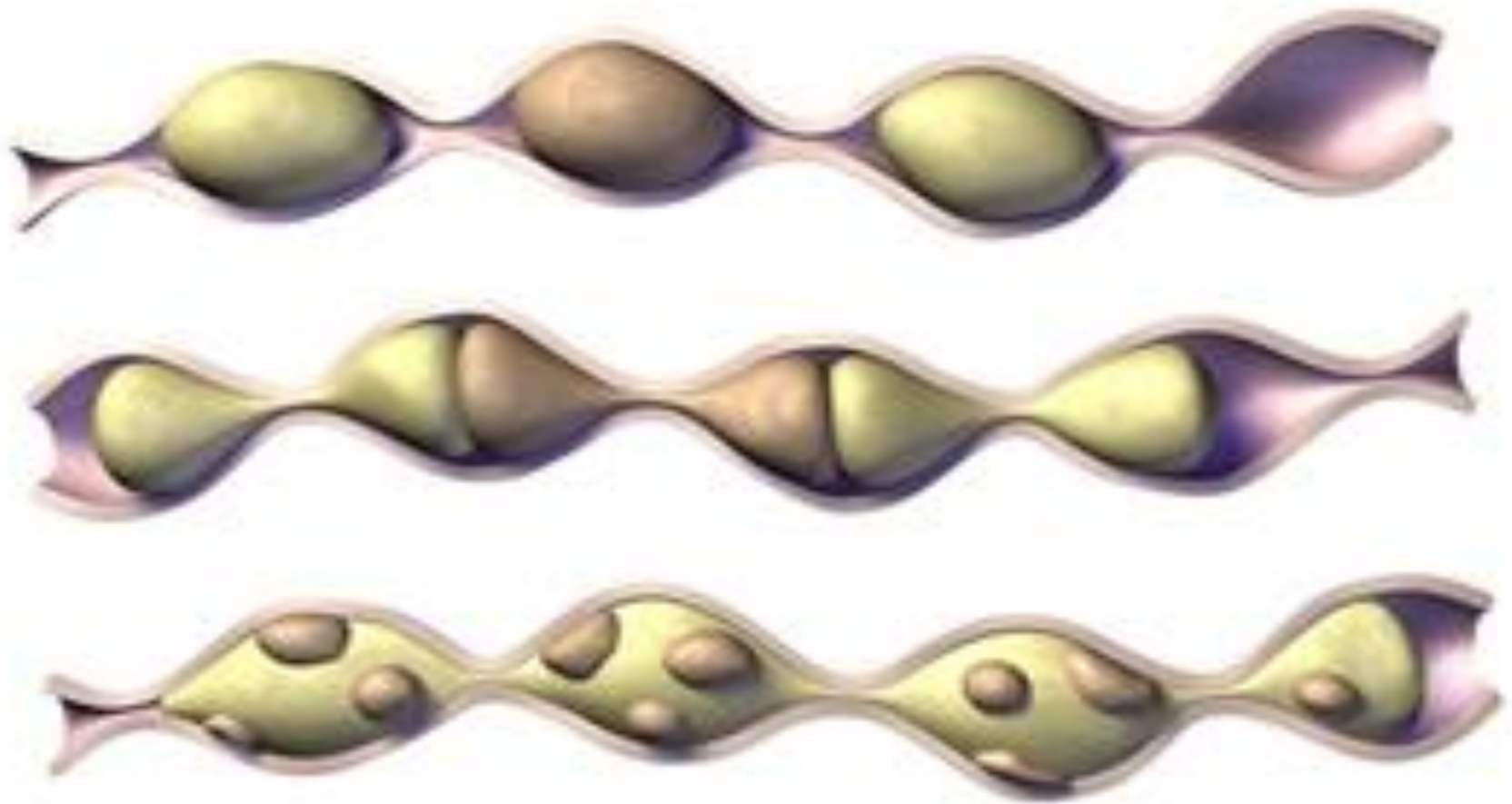
Direction of movement



Mixing movement

They are local intermittent constrictive contractions occur every few cms in the gut wall, lasts only 5 -30 seconds then new constrictions occur at other points keeping the contents thoroughly mixed together

Mixing movement



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Tonic contractions

- continuous contractions with different intensity lasts minutes to hours.**
- It is caused by continuous repetitive spike potential, any factor causes continuous partial depolarization, or by hormones.**
- It is usually occurs at the sphincteric regions**

Hunger contraction

It is rhythmical continuous tetanic intense prestalsis on empty stomach, lasts 2-3 minutes in young.

It may cause mild pain in the pit of stomach, hunger pangs especially after 12 hours – 4 days of starvation then it is gradually weakened.

Receptive relaxation of the **stomach:**

When the esophageal peristalsis approaches the stomach , a wave of relaxation occurs in the entire stomach to the duodenum to prepare to receive food.

It is due to ++ of myentric inhibitory neurons.

Reverse peristalsis

(antiperistalsis)

It is peristaltic waves occur in the reverse direction usually between stomach and duodenum and between ileum and cecum

Peristaltic rush

Irritation of small intestine results in powerful peristalsis travel the whole intestine in minutes to sweep its contents e.g., as on cases of infectious diarrhea

Haustration (segmentation of the colon)

it is simultaneous strong contraction of the circular muscles and teniae coli of the colon results in out plugging of the distal part to allow dug into and roll over the fecal material to absorb fluid and dissolved substances

Mass movements

It is peristalsis in the colon 1-3time/day.

20 cm distal to the constrictive ring loses

haustrations and contract as a unit to propel the

fecal content in mass then inhibited to be

returned after hours mediated by autonomic

fibers.

It appear after meals facilitated by gastro-colic

and ilio-colic reflexes.

Regulation of GI function

Autonomous
smooth muscle
function

pacemaker activity
electrical coupling

Neural regulation

extrinsic NS (CNS)
intrinsic NS

GI hormones

Paracrine
mediators

humoral regulation

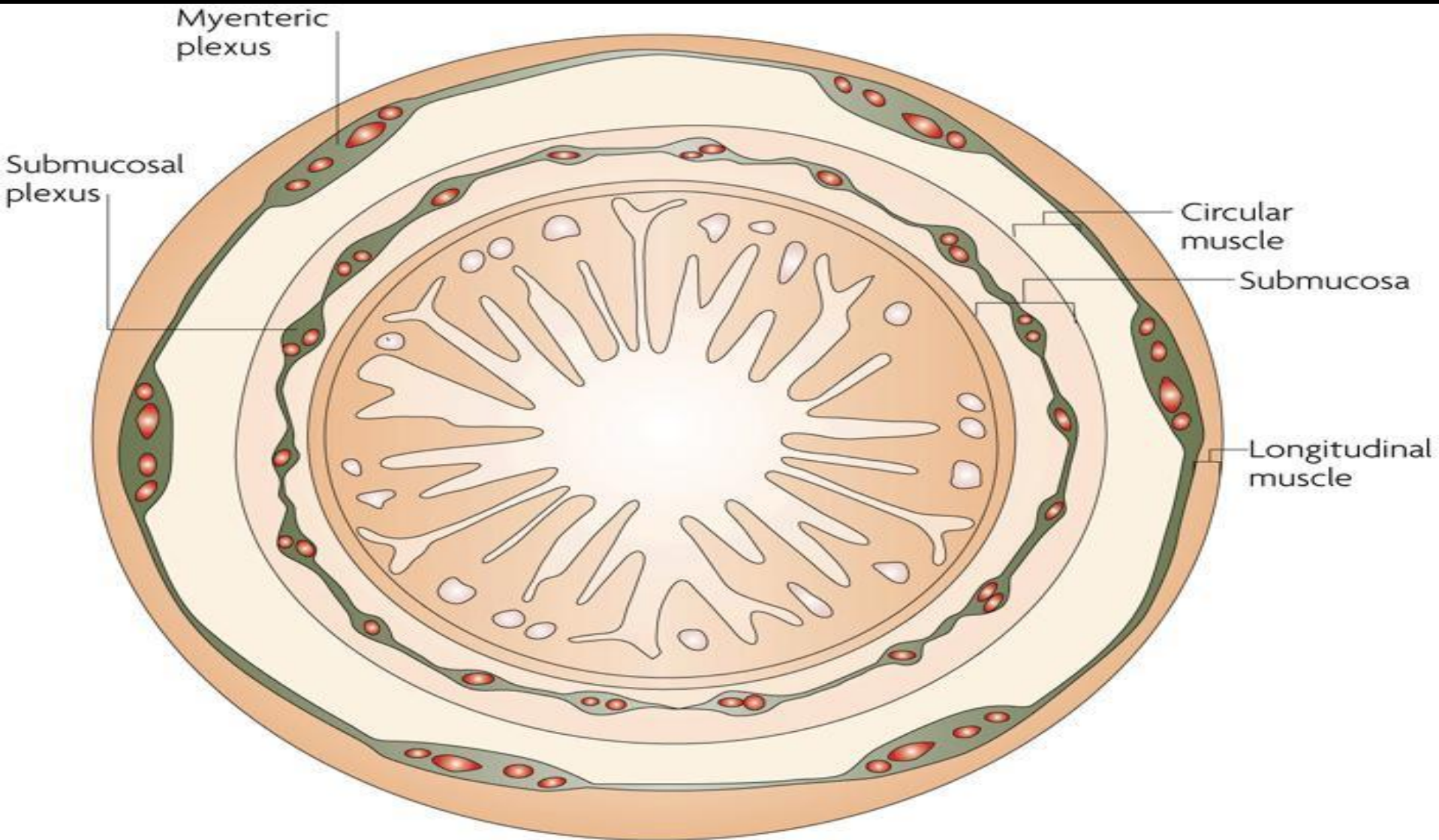
Regulation of functional components

Neural

- Enteric NS controls secretions and movement
- Comprised of 2 plexuses:
 1. Myenteric (Auerbach's)
 2. Submucosal (Meissner's)
- The plexuses are influenced by ANS

Hormonal

Enteric nervous system (gut brain)



Enteric nervous system (gut brain)

The myenteric plexus has

- ★ Sensory receptor neurons
- ★ Motor effector neurons
- ★ The neurotransmitters

Excitatory & Inhibitory

Nervous Control of the GI Tract

- Intrinsic controls
 - Nerve plexuses near the GI tract initiate short reflexes
 - Short reflexes are mediated by local enteric plexuses (gut brain)
- Extrinsic controls
 - Long reflexes arising within or outside the GI tract
 - CNS centers and extrinsic autonomic nerves

Nervous Control of the GI Tract

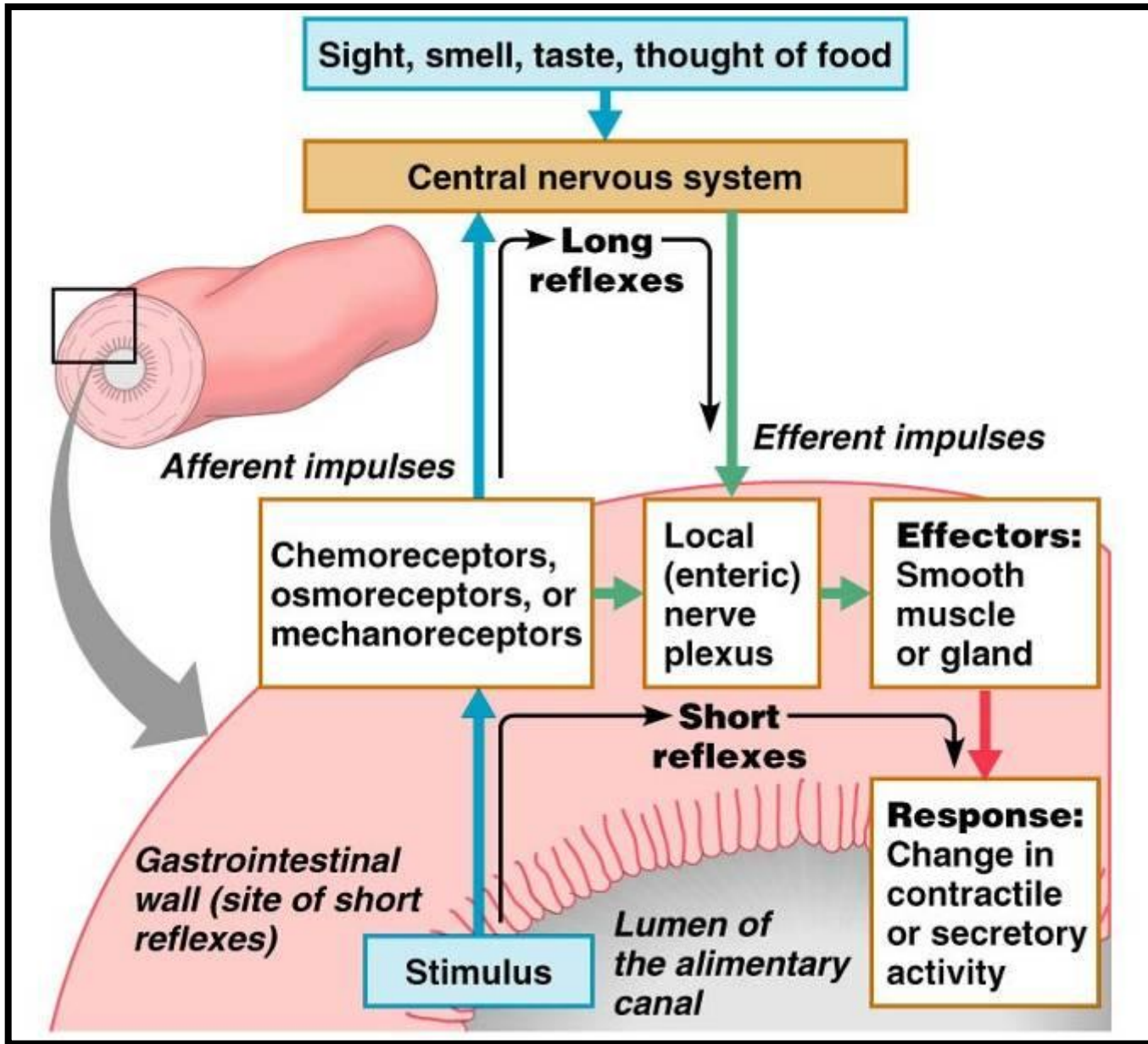


Figure 23.4

Blood Supply: Splanchnic Circulation

- Arteries and the organs they serve include
 - The hepatic, splenic, and left gastric: spleen, liver, and stomach
 - Inferior and superior mesenteric: small and large intestines
- Hepatic portal circulation:
 - Collects nutrient-rich venous blood from the digestive viscera
 - Delivers this blood to the liver for metabolic processing and storage

Blood supply to GI system

Blood Supply: Splanchnic Circulation

- Celiac trunk (hepatic, splenic, L. gastric) to spleen, liver, & stomach
- sup & inf mesenteric to sm & lg intestines
 - Celiac trunk & mesenterics receive 1/4 of C.O.
 - percentage increases after ingestion of a meal
- hepatic portal system collects nutrient rich venous blood draining from digestive viscera & takes it to the liver
 - liver collects nutrients for metabolism or storage b/f releasing them back to the bloodstream (via hepatic vein) for general cellular use